BORE—Detecting RNA Editing Events Comfortably

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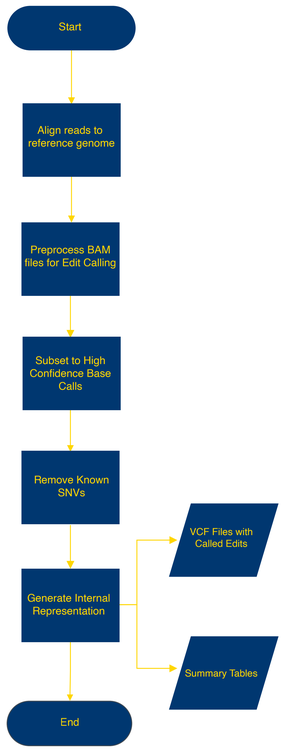
Introduction

RNA Editing is a well defined process in messenger RNA is modified post-transcriptionally. This process can result in truncation of the protein or different structures and conformations of the translated protein. Understanding and quantifying the occurrence of these processes is an important step in understanding how RNA editing influences disease processes.

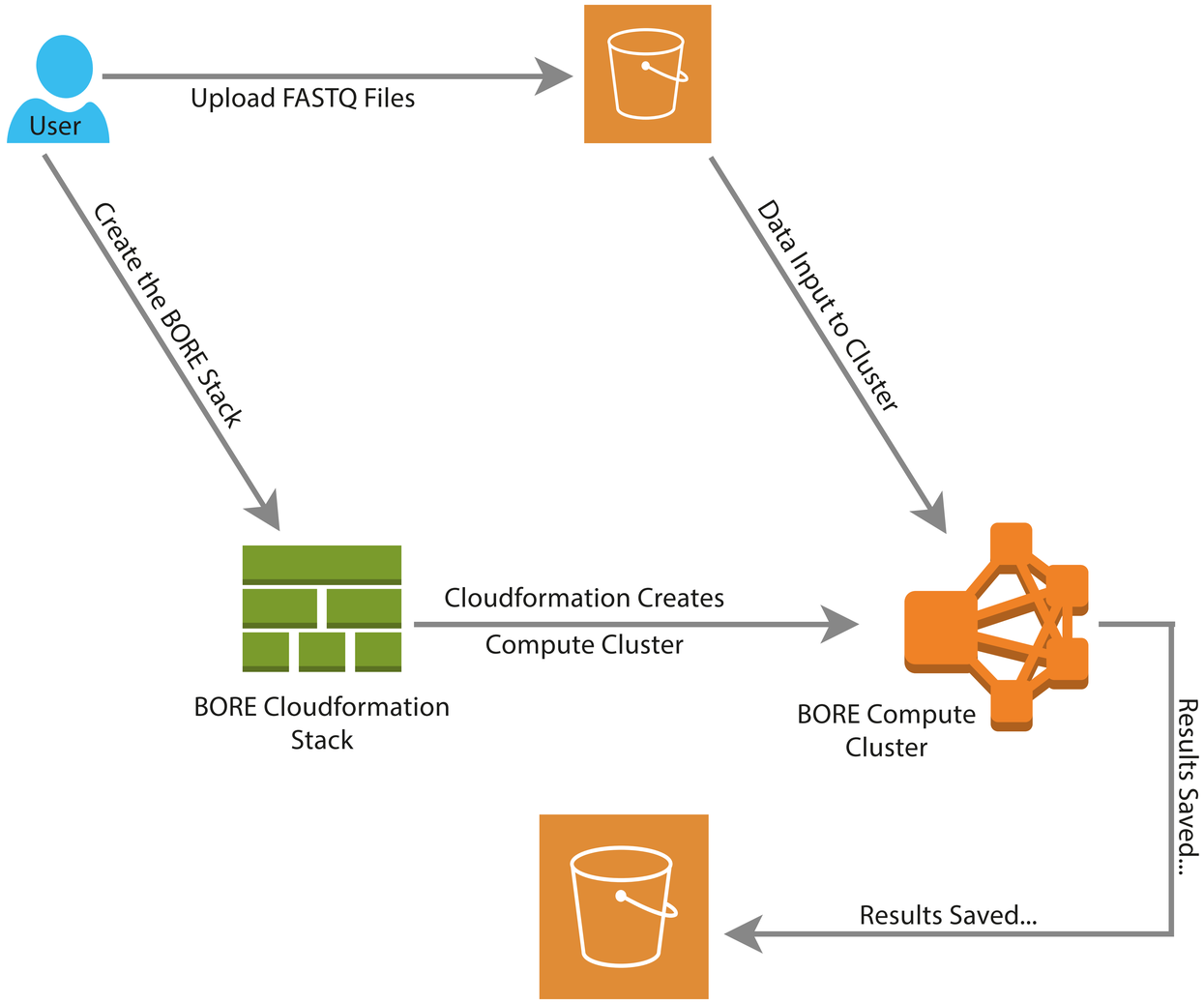
In this paper we introduce Balanced Output of RNA Editing or BORE, a pipeline developed to quickly and accurately identify potential RNA Editing sites from high-throughput sequencing data.

Bore relies on successive filtering to generate a list of potential RNA Editing sites and then outputs them in a manner that is easily usable by existing tooling. It also extracts the relative number of RNA editing events versus non-editing events and calculate the likelihood of its involvement in outcomes of interest.

Conceptual Workflow



Cloud Workflow



Materials and Methods

- HISAT2 for alignment
- GATK for Variant Discovery
- AWS for cloud services

For variant filtering, the Simons project, 1000 genomes project and International HapMap project are used identify known variants. The process is designed to run autonomously on the cloud.

Availability

The project is open-source and available on GitHub.

GitHub

Future Directions

- Generating a RADAR-equivalent database of Editome for GRCh38
- Creating a public server for people to use freely
- Better Visualization
- Connecting to UCSC Genome Browser as a custom track
- Integrating Differential Editing Analysis into the pipeline

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